

Expectation vs. Reality: Auditing Data Deletion in Commercial LLM Chatbots and What Users Believe It Does

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Abstract—Users can request the deletion of their data from LLM-based chatbot platforms, but it remains unclear whether these deletion mechanisms operate in ways that align with user expectations. We address this gap through a two-part study that combines a technical audit of data erasure practices across six commercial platforms with a user survey examining deletion expectations and behaviors.

I. INTRODUCTION

RQ: How do users delete data in AI chatbots? How are different data deletion methods evaluated?

- Establish that commercial LLM chatbots now retain personal data across sessions through memory features, making deletion a live user concern.
- Introduce the central gap between what platforms say (policy), what they do (behavior), and what users believe deletion does.
- Explain why this matters: legal erasure rights assume deletion is meaningful, but LLM memory may make it partial or effectively illusory.
- State the prior-work gap: audits measure behavior but not expectation, surveys measure perception but not behavior, and none pair them for LLM erasure.
- Present the two-part approach pairing a technical audit across six platforms with a participatory study of deletion expectations.
- Lay out the research questions split by side: system (RQ1 behavior, RQ2 policy gap), user (RQ3 expectation and verification, RQ4 sensitivity) and divergence (RQ5).
- Enumerate the contributions: first paired audit-plus-survey of RTBF in LLM chatbots, a reusable probe battery, a per-platform policy-vs-practice comparison, and a user-phrasing-to-method mapping.

II. BACKGROUND AND RELATED WORK

A. *The Right to be Forgotten and Data Deletion*

- Introduce GDPR Article 17 and the legal expectation that erasure is real and complete.
- Review prior deletion and deletion-privacy work on social platforms as the closest lineage.
- Argue that RTBF assumptions do not cleanly transfer to LLM memory, motivating the study.

B. *Privacy and Memory in LLM Systems*

- Summarize LLM memorization and data-leakage risks as the technical backdrop.
- Describe how memory and persistence features actually work in commercial chatbots.
- Review black-box privacy audits of LLM and GenAI systems and state what they measured.

C. *User Perceptions of Privacy and Deletion*

- Survey chatbot privacy-perception studies and the awareness-to-practice gap.
- Review deletion expectations and the privacy paradox.
- Position this work against its closest predecessor and name the missing paired design as the gap.

III. PARTICIPATORY STUDY

A. *Methodology*

1) *Instrument Overview:*

- Name the three measurement types the survey combines: objective knowledge, attitudes, and scenario expectations.
- Map the survey blocks to the user-side research questions.

2) *Scenario Design:*

- Describe the two fixed scenarios (Weather, low sensitivity; Health, high sensitivity) as a within-subject sensitivity manipulation.
- Explain the participant-selected deletion method piped forward into the per-method evaluation items.
- Describe the free-text elicitation of deletion phrasings and rationale, the bridge to the audit.

3) *Measures:*

- Detail the knowledge items and how they are scored.
- Detail the attitudinal Likert batteries (privacy comfort, social influence).
- Detail the scenario expectation, prior-verification, and cost-benefit rating items.

4) *Pilot and Refinement:*

- Describe the pilot round(s), what changed, and comprehension checks.

5) *Recruitment and Deployment:*

- State the Prolific deployment, screening criteria, sample-size target, and IRB approval.
- Describe the embedded attention checks and the exclusion rule.

6) *Participant Demographics:*

- Report sample composition and how it compares to a reference population.

7) *Analysis:*

- Explain the descriptive statistics and the within-subject sensitivity comparison.
- Explain the qualitative coding of free-text responses (codebook, two coders, agreement).

8) *Ethics:*

- State consent, confidentiality, and data-handling practices.

B. *Results*

1) *What Users Know About Memory and Deletion:*

- Report objective knowledge scores and the specific misconceptions that dominate.

2) *Deletion Expectations and the Sensitivity Effect:*

- Report scenario expectations and the within-subject Weather vs. Health comparison.
- Report prior-verification behavior and cost-benefit ratings by method.

3) *How Users Phrase Deletion Requests:*

- Report the coded free-text phrasings and rationales, the input to the audit’s erasure methods.

IV. TECHNICAL AUDIT

A. *Methodology*

- Define injection, erasure, and recall precisely as the paper’s core vocabulary.
- Define anchor, memory layer, and what “retained” means operationally.

B. *Recall Probe Taxonomy*

- Present the recall taxonomy: probe style (direct, settings, indirect) crossed with session (same, cross).

C. *Adversary and User Model*

- Frame the diligent-user model, a motivated user exhausting all deletion mechanisms, and define what counts as a deletion failure.

1) *Platform Selection:*

- State the six platforms and the selection rationale (coverage, popularity, memory-feature variation).
- Describe the fresh-account protocol and the Perplexity-last contamination control.
- Single out DeepSeek as the no-memory-surface case where the finding is a policy gap, not a behavioral failure.

2) *Battery Design:*

- Explain the singles-only design: one setup per (injection, erasure) cell plus one maximal combination per platform.
- Note that each platform has its own enumeration, given in full in Appendix A.

3) *Per-Cell Procedure:*

- Describe the fixed per-cell order (inject, verify, erase, probe) with the direct probe last to avoid priming.
- Explain injection verification as a mandatory second step confirming the anchor entered the intended layer.
- State the maximal-cell internal ordering and the session-destroyed rule.

4) *Platform-Specific Handling:*

- Describe the delayed memory-synthesis timing and the before-synthesis timing-variant cell.
- Describe the file-upload sub-study and its two erasure conditions per platform.

5) *Scoring and Analysis:*

- Explain scoring each outcome against a pre-registered expected outcome, with the discrepancy against the platform’s claim as the finding.
- State the rerun rule, the single-replicate baseline, and the treatment of architectural nulls.

6) *Policy Comparison Method:*

- Describe collecting each platform’s stated policy and coding observed-versus-disclosed handling.

D. *Results*

1) *Erasure Behavior Across Platforms:*

- Lead with the cross-platform erasure pattern as the headline result.
- Give per-platform detail keyed to the cell matrix.
- Highlight notable deletion failures and resurrections.

2) *Policy vs. Practice:*

- Present the per-platform policy-gap table.
- Detail cases where observed behavior contradicts stated policy.

3) *File-Upload Sub-Study Findings:*

- Report where uploaded content entered memory and survived erasure.

E. *Knowledge and Awareness*

- Report what users do and do not know about LLM data handling and RTBF.

F. Deletion Attitudes and Behavior (RQ3)

- Report self-reported deletion practices and whether users verify.
- Report what users expect deletion to actually accomplish.

G. Effect of Data Sensitivity (RQ4)

- Report the Weather-versus-Health contrast on intent, expectation, and effort tolerance.

H. Post-Deletion Feedback Expectations

- Report what confirmation or feedback users expect after deleting.

I. Elicited Deletion Phrasings

- Report how users actually phrase erasure requests, feeding the synthesis.

V. SYNTHESIS: EXPECTATION VS. REALITY

- Frame the section by placing measured platform behavior directly beside user expectation.
- Comparison A: tabulate expected deletion outcome against audited outcome, per method class.
- Comparison B: map user-elicited phrasings onto tested methods that work versus methods that fail.
- Report the divergence (RQ5): where the gap is widest and why it matters.
- Identify who bears the gap: which user groups or data types are most exposed.

VI. DISCUSSION

A. Interpreting the Gap

- Discuss what the expectation-reality distance implies for meaningful consent and user control.

B. Design Implications

- Argue for deletion confirmation and feedback, tied to the post-deletion expectations findings.
- Argue for honest memory surfaces and clear disclosure of erasure scope.

C. Policy Implications

- Discuss RTBF enforceability for LLM memory and policy-versus-practice accountability.

D. Recommendations by Stakeholder

- Give lightweight recommendations for developers, platforms, regulators, and users .

VII. LIMITATIONS

- Note the self-selected method confound created by piping the participant's chosen deletion method forward.
- Note the free-tier, fresh-account, single-replicate scope of the audit.
- Note probabilistic model behavior and the temporal validity of findings.
- Note self-report biases on the survey side.

VIII. CONCLUSION

- Restate the measured gap and its significance in one tight paragraph.

ACKNOWLEDGMENT

To be written.

APPENDIX A

FULL INJECTION/ERASURE/RECALL BATTERY

- Per-platform enumeration, per-cell reading schedule, setup counts, and file sub-study.

APPENDIX B

SURVEY INSTRUMENT

- Full questionnaire grouped by block: Screening, Usage Profile, Weather Scenario, Deletion Behaviour, Post-Deletion Feedback Expectations, Privacy Awareness, Demographics, Consent, Social Influence, Health Scenario.

APPENDIX C

PRE-REGISTRATION TABLE (OPTIONAL)

- Per-cell expected outcomes.